

DS 2500 *Fall* 2025 Lab Exercise 08

Deadline : 11/03/2025 11:59 PM EST

Late Deadline: 11/24/2025 11:59 PM EST with 50 points off

Total Points: 200 pts

Grading:

- This lab assignment is auto-graded. Gradescope can be a little picky, so make sure you don't put extraneous characters or whitespace in your answers -- and double-check the correct/incorrect confirmation.

Submission Instructions:

- Submit the Python file (.py) containing your solutions to Gradescope by the deadline specified at the top.
- You're allowed *unlimited* submissions.
- There are two deadlines. Submissions will not be accepted after the late deadline.

Disclosure: We've adapted lab problems from DS 2500 Spring 2025, so one or more problems may be the same or similar to those from that course.

Setup Code

- Use the following code to begin this lab

```
import pandas as pd
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
from sklearn.model_selection import train_test_split

df = pd.read_csv("lab08_data.csv")
```

Problem 1 (50 pts)

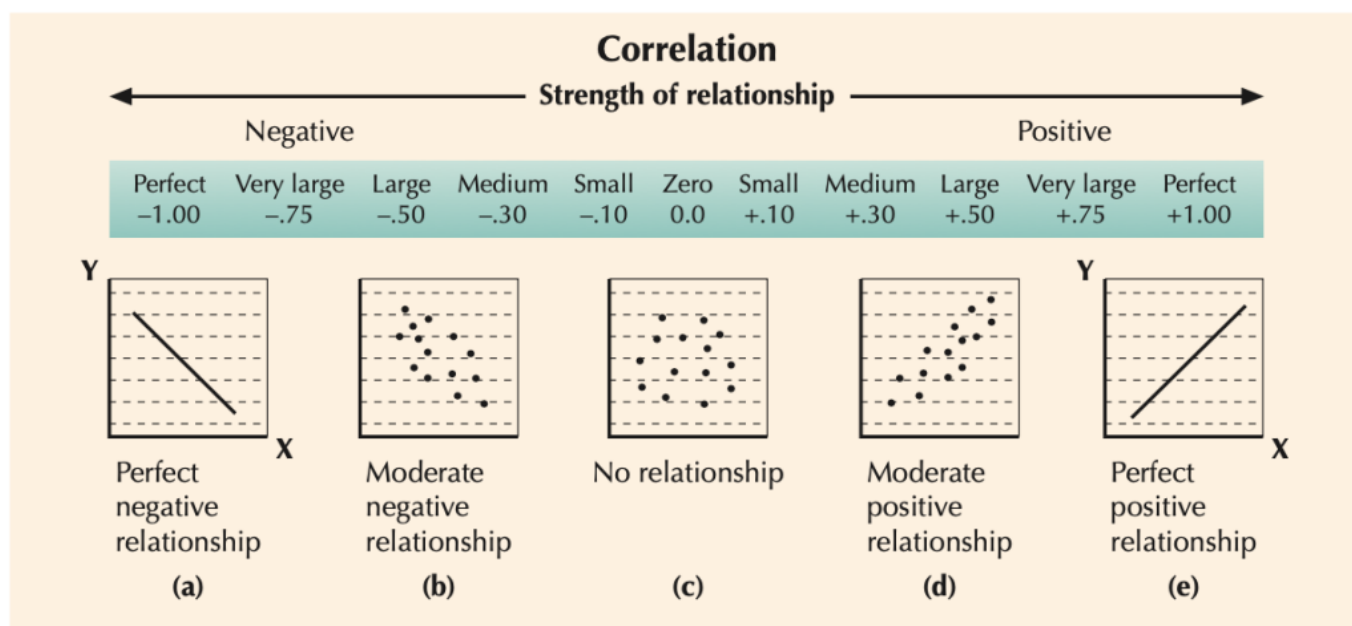
- Use `train_test_split()` to split the dataframe into 2 dataframes
 - `test_size` must be set to `0.2`.
 - `random_state` must be set to `2500`.
- The resulting dataframes should be named as `df_tr` and `df_te`.
- Hint: You can pass the entire dataframe `df` directly into `train_test_split()` to get the split.

Problem 2 (50 pts)

Pearson's Correlation is a statistical measure that indicates the extent to which two variables change together, or are related. It is measured by a value called the correlation coefficient, which ranges from -1

(perfect negative correlation) to +1 (perfect positive correlation), with 0 indicating no linear correlation.

See Figure below.



The Pandas DataFrame has a method called `.corr()` that computes pairwise correlations between numeric columns and displays them as a matrix.

For Problem 02, run the following code and see how much each feature is correlated with y:

```
corr_matrix = df_tr.corr()
```

Problem 3 (50 pts)

- Use `LinearRegression()` to instantiate a model
- Train the model using one feature from the training set.
 - Look at the correlation matrix and use the feature that is most correlated with y.
 - Hint: `model.fit()`
- Store the weight (slope) into the variable `w`.
- Store the bias (intercept) into the variable `b`.

Problem 4 (50 pts)

- Use `mean_squared_error()` and `r2_score()` to do the followings:
- Make predictions using the training set
 - Hint: `model.predict()`
 - Compute the evaluation metrics
 - Store the mean squared error value into the variable `mse_tr`.
 - Store the r^2 value into the variable `r2_tr`.
- Make predictions using the test set
 - Compute the evaluation metrics
 - Store the mean squared error value into the variable `mse_te`.

- Store the r^2 value into the variable `r2_te`.